

CSC 1012

Introduction to Computer Programming

Individual Assignment

Logistic Management System

In C language

Student Name: M.K.P.Ransaka

Index No: AS20240565

❖ Introduction

This project is a simple C program designed to manage and calculate delivery operations between different cities. It allows users to add city names, set distances, and select a vehicle type to estimate delivery cost, fuel usage, profit, and time. The system uses basic programming concepts such as arrays, loops, conditionals, and functions to perform calculations and display results in a clear, menu-driven format.

❖ Objectives

1. To design a simple **menu-driven logistics management system** using the C programming language.
2. To allow users to **add, rename, and remove cities** in a list.
3. To **store and edit distances** between cities using a 2D array.
4. To **select vehicle types** (Van, Truck, Lorry) with different capacities and rates.
5. To **calculate delivery cost, fuel cost, profit, and delivery time** based on user input.
6. To **validate user inputs** such as vehicle capacity and city selection.
7. To demonstrate the use of **arrays, loops, functions, and conditionals** in solving a real-world problem.

❖ System Features

- Add, rename, and remove cities
- Store up to 30 cities
- Enter and display distances between cities using a 2D array
- Choose a vehicle type: Van, Truck, or Lorry

- Input weight and validate vehicle capacity
- Calculate:
 - ✓ Base delivery cost
 - ✓ Fuel usage and fuel cost
 - ✓ Total operational cost
 - ✓ Profit and customer charge
 - ✓ Estimated delivery time
- Display all calculated results clearly in the console

❖ Project Design Overview

The system is designed using a menu-driven structure with separate functions for each main operation:

1. City Management:

Handles adding, renaming, and removing city names stored in an array.

2. Distance Management:

Uses a 2D array to store distances between cities. The program ensures distances are symmetrical.

3. Delivery Handling:

Collects source, destination, vehicle type, and weight.

Performs all delivery-related calculations using predefined formulas.

Data such as vehicle capacity, rate per kilometer, speed, and efficiency are stored in simple arrays. The program continues running until the user chooses to exit.

❖ Key Calculations

$$\text{Cost} = D \times R \times (1 + W / 10000)$$

$$\text{Fuel Used} = D / E$$

$$\text{Fuel Cost} = \text{Fuel Used} \times 310$$

$$\text{Total Cost} = \text{Cost} + \text{Fuel Cost}$$

$$\text{Profit} = 0.25 \times \text{Cost}$$

$$\text{CustomerCharge} = \text{TotalCost} + \text{Profit}$$

$$\text{Time} = D / S$$

❖ Example Output

From: Kandy

To: Galle

Vehicle: Van

Weight: 500 kg

Distance: 120 km

Base Cost: 3660.00 LKR

❖ Limitations

- No file saving/loading features
- Distances must be entered manually each time
- No automatic route optimization

❖ Conclusion

The Logistics Management System is a simple yet effective C program that demonstrates how programming concepts can be applied to solve real-world problems. By using arrays, loops, and functions, this system successfully simulates a small-scale logistics operation that manages cities, calculates delivery costs, and estimates travel time. It provides a practical understanding of how structured programming can be used to build functional and user-friendly applications.

❖ GitHub Repository

URL: <https://github.com/Soulrender268451/logestic-management-c.git>